

Harsh Bari

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 30 April 2001 |  Moti Daman |  harshbari-153.github.io (Portfolio)

PROJECTS

Autonomous AI Agent for EDA | GEMMA 4 26B, Nvidia T4 (x2) GPU, vLLM May 2026

- Takes a CSV file and generates 70-100 insights for Regression, Classification, Clustering, Association.
- Specific handling of Outliers, Feature Analysis/Transformation, Hyperparameter Tuning, Ensembling.
- Demo Video: <https://youtu.be/7pGsFObpbm8>

Math Olympiad AI Agent | H100 GPU, GPT-OSS-120B (MXFP4) March 2026

- AI Agent solving Algebra, Geometry, Combinatorics & Number Theory, running on vLLM.
- Deterministic tool calling using OpenAI Harmony Response Format and weighted entropy based voting.
- Solved 4/4 easy, 3/4 moderate and 1/2 hard questions from AIMO3 Reference Benchmark Questions.

Next Assignment | PostgreSQL, Gemini-2.5-Flash, Streamlit, S-Transformers Sept 2025

- Generates real-world assignments from trending Indian news from user given subject and skills.
- Automated ETL pipeline fetching top 5 Indian news daily at 12 AM via GNews API, and stores results on Neon Website using GitHub Actions for CI/CD pipeline with unit testing and dependency setup.
- Deployed at: next-assignment.streamlit.app
- Demo Video: bit.ly/3KcqPRy

KJNN | PyTorch, BeautifulSoup, ScikitLearn, ML, NLTK, Keras, Gensim Jan 2025

- Neural Network Architecture which classifies Claim-Justification Sentences into 6 classes.
- Trained 7 matrices(6 100x100, 1 200x100) on Politifact Dataset (21k news articles) scraped from Kaggle.
- Outperforms sBERT, DistilBERT, GloVe, RoBERTa-Base, and Word2Vec in 3/5 ANN test classifiers.

EDUCATION

Aug 2023 - Sept 2025	MTech Data Science at NIT Surat	7.88 CGPA
July 2019 - March 2023	BTech CSE at PIET Vadodara, Gujarat	8.70 CGPA
July 2017 - May 2019	Class 12th, Gujarat Board at IOLF Daman	79.93%
July 2016 - May 2017	Class 10th, Gujarat Board at IOLF Daman	80.67%

PUBLICATIONS

Generalized Immutable Ledger (GILED) using Blockchain Technology	IEEE SCEECs'23
Available at: ieeexplore.ieee.org/document/10062983	March 2023
Contextual Sentence Embedding Base Difference Procedure (BDP) Vectorization	IEEE SCEECs'25
Available at: ieeexplore.ieee.org/document/10940513	March 2025
Claim Justification using Subject Predicate Distance (SPD) Extraction	IEEE ICDSBS'2025
Available at: ieeexplore.ieee.org/document/11031662	June 2025
Claim Justification Contextual Embeddings using KJNN	IEEE IC2E3'2025
Accepted on 8 April 2025(paper withdrawn)	May 2025

WORK EXPERIENCE

Executive Research & Development Department	1 Nov 2025 - Present
Robotics workflow using Leerobot repository, Isaacsim, IsaacLab.	Vapi, Gujarat
Part Time Teaching Assistant (Taking Lectures and Labs)	Nov 2021 - Oct 2022
Taught Java, Data Structures in 2nd, 3rd year BTech with 90% positive feedback.	PIET Vadodara, Gujarat
Subject Matter Expert at Chegg	Feb 2021 - Dec 2022
Posted 500+ solutions on Chegg Consulting Website in Computer Science.	Chegg Website

OTHERS

Languages:	Gujarati, Hindi, English.
Achievements:	GATE CSE 2023 (AIR 3251), CS UGC NET Jan-2025 (97%ile).
Skills	hugging face, web scraping, local llm orchestration on vLLM server.
Problem Solving:	300+ questions on geeksforgeeks.org/profile/bariharsh2001?tab=activity